58. Who is not a Nobel Prize winner among the following scientists ?

(a) Har Govind Khorana (b) C.V. Raman
(c) S. Chandrashekhar (d) Jagdish Chandra Bose

U.P.P.C.S. (Mains) 2006

Ans. (d)

Dr. Har Gobind Khorana was awarded Nobel Prize in Physiology or medicine in 1968. Sir Chandrasekhara Venkata Raman was awarded Nobel Prize in Physics for his work on the scattering of light in 1930. Subrahmanyan Chandrasekhar was awarded Nobel Prize in physics in 1983. Jagdish Chandra Bose is not a Noble Prize winner.

59. 2023 Nobel Prize in Physics awarded for :

(a) Experimental methods that generate attosecond pulses of light for the study of electron dynamics in matter
(b) Experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science
(c) Physical modelling of Earth's climate, quantifying variability and reliably predicting global warming
(d) None of the above

Chhattisgarh P.C.S. (Pre) 2023

Ans. (a)

The Nobel Prize in Physics, 2023 was awarded to Pierre Agostini, Ferenc Krausz and Anne L'Huillier "for experimental methods that generate attosecond pulses of light for the study of electron dynamics in matter." The Nobel Prize in Physics, 2024 was awarded jointly to John J. Hopfield and Geoffrey Hinton "for foundational discoveries and inventions that enable machine learning with artificial neural networks."

60. Who is the recipient of the 2023 International Prize in Statistics equivalent to the Nobel Prize in Mathematics?

(a) David R. Cox
(b) Bradley Efron
(c) Calyampudi Radhakrishna Rao
(d) Nan Laird

M.P. P.C.S. (Pre) 2024

Ans. (c)

The 2023 International Prize in Statistics, often considered the 'Nobel Prize' of statistics, was awarded to Calyampudi Radhakrishna Rao (C.R. Rao), an Indian-American mathematician and statistician, for his contributions to the field of statistics. The International Prize in Statistics is awarded every two years to an individual or team "for major achievements using statistics to advance science, technology and human welfare."

Miscellaneous

Notes

Indian Science Congress Association (ISCA) :

- The Indian Science Congress Association is a professional body under Department of Science & Technology, Government of India.
- The ISCA is a premier scientific organization of the country established in 1914, with headquarters at Kolkata (W.B.). It owes its origin to the foresight and initiative of two British Chemists, namely, Prof. J.L. Simonsen and Prof. P.S. MacMahon.
- ISCA has been promoting science and inculcating the spirit of science through its multifarious activities.
- The ISCA is a society registered under Societies Act XXI of 1860.
- The ISCA vision is to inculcate scientific temper among the common people.
- The ISCA mission is to make people aware about the recent developments in science and technology and its impact on the society, to establish different centres all over India for spreading scientific knowledge through school / college programs, to provide a common platform where scientists from India and abroad can exchange their views.
- The first Indian Science Congress was held in 1914 at the premises of the Asiatic Society, Calcutta (Now Kolkata).
- The 104^{th} ISC conference was held at Sri Venkateshwar University, Tirupati, (A.P.) from 3-7 Jan 2017; 105^{th} conference held at Manipur University (Imphal) from 16-20 March 2018; 106^{th} conference was held at Lovely Professional University, Phagwara, Jalandhar, Punjab from 3-7 Jan. 2019 and its focal theme was- 'Future India : Science &Technology'.
- Its 107^{th} conference was held at University of Agricultural Sciences, GKVK Campus, Bengaluru with the focal theme of 'Science & Technology : Rural Development' from 3-7 Jan. 2020.
- Due to COVID-19 pandemic annual ISC conference was cancelled in 2021 and 2022.
- 108^{th} ISC conference was held from 3-7 January, 2023 at Rashtrasant Tukadoji Maharaj (R.T.M.) Nagpur University with the focal theme of 'Science and Technology for Sustainable Development with Women Empowerment'.
- The 109^{th} edition of the ISC was originally meant to hosted by Lucknow University (LU) on the theme 'Global Perspective on Science and Technology for a Sustainable Future'. However, LU withdrew from the responsibility and the venue was shifted to Lovely Professional University (LPU), Jalandhar but LPU has also backed

out from hosting this event. Ultimately it was not held due to a dispute between the Department of Science and Technology (DST) and the Indian Science Congress Association (ISCA).

Nanotechnology :

- The branch of technology that deals with the dimensions and tolerances of less than 100 nanometres, especially the manipulation of individual atoms and molecules.
- Nanometre is a unit of length (1 nanometre = 10^{-9} metre)
- The matter on nanoscale is too small which can not be seen by the naked eyes, even though they are invisible in microscope.
- To observe matter on nanoscale Scanning Tunneling Microscope or Atomic Force Microscope is used.
- Eminent physicist Richard Feynman firstly expressed his views on this subject at California Technology Center on 29 Dec. 1959 and initiated an era of nanotechnology.
- The term nanotechnology was firstly defined by Tokyo Science University Professor Norio Taniguchi in 1974. According to him nanotechnology mainly consists of the processing, separation, consolidation and determination of materials by one atom or by one molecule.
- In 1986, Dr. K. Eric Drexler wrote a book on nanotechnology – 'Engines of Creation: The coming era of Nanotechnology', which is the first book on this subject.

LASER :

- A laser is a device that emits light through a process of optical amplification based on the stimulated emission of electromagnetic radiation.
- The term LASER originated as an acronym for "Light Amplification by Stimulated Emission of Radiation".
- The emission generally covers an extremely limited range of visible, infrared or ultraviolet wavelengths.
- The main peculiar feature of laser is that frequency, amplitudes and polarization is same. Minimum scattering takes place, so total energy is centralized to a point.
- Max Planck Institute of Quantum Optics (Munich, Germany) is a leading institute in the world regarding fundamental researches on laser.

MASER :

- A maser is a device using the stimulated emission of radiation by excited atoms to amplify or generate coherent monochromatic electromagnetic radiation in the microwave range.
- MASER is an acronym for "Microwave Amplification by Stimulated Emission of Radiation."
- The first maser was built by Charles H. Townes, James P. Gordon, and Herbert J. Zeiger at Columbia University in 1953.
- Masers are used as the timekeeping device in atomic clocks, and as extremely low-noise microwave amplifiers in radio telescopes and deep space spacecraft communication ground stations.
- Modern masers can be designed to generate electromagnetic waves at not only microwave frequencies but also radio and infrared frequencies. For this reason Charles Townes suggested replacing "microwave" with the word "molecular" as the first word in the acronym maser.
- The laser works by the same principle as the maser, but produces higher frequency coherent radiation at visible wavelengths.

Liquid Crystal :

- A liquid crystal is a thermodynamic stable phase characterized by anisotropy of properties without the existence of a three-dimensional crystal lattice, generally lying in the temperature range between the solid and isotropic liquid phase, hence the term mesophase.
- Liquid crystal materials are unique in their properties and uses. As research into this field continues and as new applications are developed, liquid crystals will play an important role in modern technology.
- Liquid crystal materials generally have several common characteristics. Among these are a rod like molecular structure, rigidness of the long axis, and strong dipole and/or easily polarizable substituents.
- Liquid crystals find wide use in liquid crystal displays, which rely on the optical properties of certain liquid crystalline substances in the presence or absence of an electric field.
- Liquid crystals have a multitude of other uses. They are used for nondestructive mechanical testing of materials under stress. This technique is also used for the visualization of RF (radio frequency) waves in waveguides. They are used in medical applications where, for example, transient pressure transmitted by a walking foot on the ground is measured. Low molar mass (LMM) liquid crystals have applications including erasable optical disks, full color "electronic slides" for computer-aided drawing (CAD), and light modulators for color electronic imaging.

Indian Antarctic Programme :

- The Indian Antarctic Programme is a multi-disciplinary, multi-institutional programme under the control of the National Centre for Polar and Ocean Research (Goa), Ministry of Earth Sciences, Government of India.
- It was initiated in 1981 with the first Indian expedition to Antarctica.
- The programme gained global acceptance with India's signing of the Antarctic Treaty and subsequent construction

of the Dakshin Gangotri Antarctic research base in 1983-84, superseded by the Maitri base from 1989.

- The newest base commissioned in 2012 is Bharati, constructed out of 134 shipping containers.
- Under the Antarctic programme, atmospheric, biological, earth, chemical and medical sciences are studied by India.

Indian Arctic Programme :

- The Indian Arctic Programme is also under the control of the National Centre for Polar and Ocean Research (Goa), Ministry of Earth Sciences, Government of India.
- India launched its first scientific expedition to the Arctic Ocean in 2007 and opened a research base named "Himadri" at the International Arctic Research Base at Ny-Alesund, Svalbard, Norway in July, 2008 for carrying out studies in disciplines like Glaciology, Atmospheric sciences & Biological sciences.
- IndARC is India's **first multi-sensor underwater moored observatory in the Arctic region.** It was deployed in 2014 at Kongsfjorden fjord, Svalbard, Norway which is midway between Norway and North Pole. Its research goal is to study the Arctic climate and its influence on the monsoon.

Question Bank

1. The branch of Physics that deals with the motion of very small particles is called :

(a) Field Theory (b) Particle Physics
(c) Quantum Mechanics (d) Atomic Physics

R.A.S./R.T.S.(Pre) 2003

Ans. (c)

Quantum Mechanics is the branch of physics that deals with the motion of very small particles.

2. Geodesy is the science that deals with :

(a) dating of terrestrial rock
(b) measurement of dimension of the Earth
(c) measurement of elevation and depression of the Earth
(d) recording of the changes undergone by the crust
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (e)

Geodesy is the science of accurately measuring and understanding three fundamental properties of the Earth : its geometric shape, its orientation in space, and its gravity field – as well as the changes of these properties with time.

3. The field of science which studies automation and communication between human and machine is called-

(a) Hydroponics (b) Cryogenics
(c) Dietetics (d) Cybernetics

R.A.S./R.T.S. (Pre) 1999

Ans. (d)

Cybernetics is relevant to study of systems, such as mechanical, physical, biological, cognitive and social systems. Cybernetics is applicable, where action by the system generates some change in its environment and that change is reflected in that system. Norbert Wiener defined Cybernetics in 1948 as "the scientific study of control and communication in the animal and machine".

4. Science of time measurement is :

(a) Horology (b) Cosmology
(c) Tomography (d) Hydrology

R.A.S./R.T.S. (Pre) 1999

Ans. (a)

Horology is the art or science of measuring time.

5. The study of friction and lubrication is –

(a) Cryogenics (b) Selenology
(c) Horology (d) Tribology

R.A.S./R.T.S. (Pre) 1999

Ans. (d)

Tribology is a branch of Mechanical Engineering and Material Science. It includes the study and application of the principles of friction and lubrication.

6. What are true about plasma state?

(i) Particles in this state are in ionic state.
(ii) Light in the LED bulb is related to plasma state.
(iii) Sunlight is also an example of plasma state.

(a) (i), (ii) and (iii) (b) (i) and (ii)
(c) (i) and (iii) (d) None of the above

Chhattisgarh P.C.S. (Pre) 2022

Ans. (c)

Plasma is often called 'the fourth state of matter', along with solid, liquid and gas, characterized by the presence of a significant portion of charged particles in any combination of ions or electrons. It is a state of matter that is created when a gas is heated to extremely high temperatures. Hence, statement (i) is true.
It is the most abundant form of ordinary matter in the universe, mostly in stars (including the Sun). Hence, statement (iii) is also true. Statement (ii) is incorrect as light in the LED bulb is not related to plasma state.

7. In which field of science, we will learn about White Dwarf?

(a) Astronomy (b) Agriculture
(c) Genetics (d) Anthropology

R.A.S./R.T.S. (Pre) 2003

Ans. (a)

'White Dwarf' is related to astronomy. It is also known as Degenerate Dwarf.

8. **Which one of the following statements is incorrect?**
(a) The special rubber tyres of aircraft are made slightly conducting.
(b) The blue waves scatter more than violet waves of light so that the sky appears blue not violet.
(c) A comb run through one's wet hair does not attract small bits of paper.
(d) Vehicles carrying inflammable material usually have metallic ropes touching the ground.
R.A.S./R.T.S. (Pre) 2008

Ans. (b)

The violet waves scattered more than blue waves of light. The main cause of sky appearing blue is the highest scattering of violet, indigo and blue colours. Rest options are correct.

9. **National Physical Laboratory is situated in –**
(a) New Delhi (b) Chennai
(c) Bengaluru (d) Kolkata
U.P. Lower Sub. (Pre) 2009

Ans. (a)

National Physical Laboratory is situated in New Delhi.

10. **National Chemical Laboratory is situated at :**
(a) Lucknow (b) New Delhi
(c) Pune (d) Hyderabad
U.P. Lower Sub. (Mains) 2015

Ans. (c)

The National Chemical Laboratory (CSIR-NCL) Pune was established in 1950 is a constituent laboratory of the Council of Scientific and Industrial Research (CSIR). CSIR-NCL is a science and knowledge-based research, development and consulting organization.

11. **Tata Institute of Fundamental research is located in :**
(a) Bangalore (b) Kolkata
(c) Delhi (d) Mumbai
R.A.S./R.T.S. (Pre) 2003

Ans. (d)

"It is the duty of people like us to stay in our own country and build up outstanding schools of research such as some other countries are fortunate to process." This was the vision that guided the Tata Institute of Fundamental Research which Homi Bhabha founded on 1st June, 1945 with the support from the Sir Dorabji Tata. It is located in Mumbai.

12. **Full form of LASER is –**
(a) Log Amplification by stimulated emission of radiation.
(b) Light amplification by stimulated emission of radiation.
(c) Locally amplified by stimulated emission of radiation.
(d) light amplification by stimulated emission of radio.
Jharkhand P.C.S. (Pre) 2003

Ans. (b)

The full form of LASER is "Light Amplification by Stimulated Emission of Radiation". A laser is a device that emits light through a process of optical amplification based on the stimulated emission of electromagnetic radiation. The first LASER was built in 1960 by the Russian scientist N.S. Basov and A.M. Prokhorov. The theoretical base came from C.H. Towres in 1965. These three physicians got the Nobel Prize for their works.

13. **Laser is a device for producing –**
(a) Spontaneous radiation (b) Dispersed radiation
(c) Scattered radiation (d) Stimulated radiation
U.P.P.C.S. (Pre) 2012

Ans. (d)

See the explanation of above question.

14. **With reference to the first-man made Ruby Laser, which of the following statement/s is/are correct?**
1. Ruby Laser produced pulses of blue light.
2. Ruby Laser produced pulses of red light.
Select the correct answer using the code given below :
(a) Neither 1 nor 2 (b) Only 2
(c) Both 1 and 2 (d) Only 1
U.P. R.O./A.R.O. (Pre) 2023

Ans. (b)

A ruby laser is a solid-state laser that uses a synthetic ruby crystal as its gain medium. The first working laser was a ruby laser made by Theodore H. Maiman at Hughes Research Laboratories (California, USA) in 1960. Ruby lasers produce pulses of coherent visible light at a wavelength of 694.3 nm, which is a deep red colour.

15. **Who is the author of 'Nuclear Reactor Time Bomb' :**
(a) C.C. Park (b) E. P. Odum
(c) S.. Polasky (d) Takashi Hirose
U.P.P.C.S. (Pre) 2011

Ans. (d)

Takashi Hirose is the author of the Book 'Nuclear Reactor Time Bomb' which he wrote in 2010.

16. **Match list-I with List- II and select the correct answer using the codes given below the list:**

List- I	List - II
A. Revolver	**1. Alfred Nobel**
B. Dynamite	**2. Pascal**
C. Law of Cooling	**3. Colt**
D. Law of pressure	**4. Newton**

Code :

	A	B	C	D
(a)	1	3	2	4
(b)	1	3	4	2
(c)	3	1	2	4
(d)	3	1	4	2

U.P. Lower Sub. (Pre) 2009

Ans. (d)

The correctly matched lists are as follows :

Revolver	–	Samuel Colt
Dynamite	–	Alfred Nobel
Law of Cooling	–	Newton
Law of Pressure	–	Pascal

17. Which of the following pair is used to form the terminals of normal torch cell?

(a) Zinc – Carbon
(b) Copper – Zinc
(c) Zinc – Cadmium
(d) Carbon – Copper

U.P.P.C.S. (Mains) 2010

Ans. (a)

Generally, dry cells are used in the torch, whose anode is made up of zinc while the cathode is made up of carbon.

18. A.T.M. means –

(a) Automatic Transaction Machine
(b) Automatic Transfer Machine
(c) Automated Teller Machine
(d) Advance Transaction Machine

U.P.R.O/A.R.O. (Mains) 2014

Ans. (c)

ATM is short for Automated Teller Machine. Basically, the ATM is used to perform banking transactions like withdrawal of money and to view bank statement.

19. Which one of the following links all the ATMs in India?

(a) Indian Banks' Association
(b) National Securities Depository Limited
(c) National Payments Corporation of India
(d) Reserve Bank of India

I.A.S. (Pre) 2018

Ans. (c)

National Financial Switch (NFS) is the largest network of shared Automated Teller Machines (ATMs) in India. It was designed, developed and deployed by the Institute for Development and Research in Banking Technology (IDRBT) in 2004, with the goal of interconnecting the ATMs in the country and facilitating convenience banking. It is run by the National Payments Corporation of India (NPCI). NPCI taken over the operations of National Financial Switch (NFS) from the IDRBT in 2009.

20. Which technology of the 21st century can do wonders in device miniaturization ?

(a) Atomic laser technique (b) Nanotechnology
(c) Gynecology (d) Hydroponics

U.P.P.C.S. (Mains) 2004

Ans. (b)

Nanotechnology is science, engineering, and technology conducted at the nanoscale which is about 1 to 100 nanometres. 1 nanometre = 10^{-9} metre. With the help of nanotechnology, we can easily make micro- accessories and equipments.

21. A particle having at least one dimension less than 10^{-7} metre, is known as

(a) Micro particle (b) Milli particle
(c) Nano particle (d) Macro particle

R.A.S./R.T.S. (Pre) 2018

Ans. (c)

Among the above options, nanoparticles are particles having at least one dimension less than 10^{-7} metre. According to the definition, a nano-object have one of its characteristic dimensions to be in the range 1-100 nm (10^{-9} – 10^{-7}m) to be classified as a nanoparticle.

22. The size of the nanoparticle ranges between :

(a) 100 nm to 1000 nm (b) 0.1 nm to 1 nm
(c) 1 nm to 100 nm (d) 0.01 nm to 0.1 nm

Uttarakhand P.C.S. (Pre) 2012

Ans. (c)

See the explanation of above question.

23. Consider the following statements :

1. Other than those made by humans, nanoparticles do not exist in nature.
2. Nanoparticles of some metallic oxides are used in the manufacture of some cosmetics.
3. Nanoparticles of some commercial products which enter the environment are unsafe for humans.

Which of the statements given above is/are correct?

(a) 1 only (b) 3 only
(c) 1 and 2 (d) 2 and 3

I.A.S. (Pre) 2022

Ans. (d)

Nanoparticles (NPs) are tiny materials having size ranging from 1 to 100 nm. Many of these are made by humans but they are also naturally produced by many cosmological, geological, meteorological, and biological processes. A significant fraction (by number, if not by mass) of interplanetary dust, that is still falling on the Earth at the rate

of thousands of tonnes per year, is in the nanoparticle range; and the same is true of atmospheric dust particles. Many viruses also have diameters in the nanoparticle range. Hence, statement 1 is incorrect.
Metal oxides in nanoparticle form such as zinc oxide and titanium dioxide now appear on the ingredient lists of household products as common and diverse as cosmetics, sunscreens, toothpastes, and medicines. Hence, statement 2 is correct.
NPs of some commercial products which enter the environment are unsafe for humans. Inhaling certain nano-sized particles may result in local lung inflammation, allergic responses or harmful effects on genes. Some particles may enter the bloodstream and accumulate in organs like the liver and spleen. Nanoparticulate matter is able to enter cells, which might in turn lead to direct and indirect genotoxic effects. Hence, statement 3 is correct.

24. 'Nano plug' refers to :

(a) A small bullet (b) A small hearing aid
(c) A small rocket launcher (d) None of the above

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (b)

'Nano plug' refers to a small hearing aid. It is so tiny that it is almost undetectable. Nano plug is comprised of micro-components and a nano-battery. It can be programmed by using software running on a computer, the result of which can be downloaded directly to the device via a cable.

25. Which one of the following statements is correct?

(a) Nanoear can detect sound levels as low as – 120 dB.
(b) Nanoear can detect sound as low as – 60 dB.
(c) Nanoear consists of a single silica nanoparticle.
(d) Nonoear consists of single silver nanoparticle.

R.A.S./R.T.S. (Pre) 2013

Ans. (b)

In 2012, the scientists of Munich University, Germany developed the first-ever 'nanoear' capable of detecting sound on microscopic length scale with an estimated sensitivity that is six orders of magnitude below the threshold of human hearing. The device is based on an optically trapped gold nanoparticle and it can detect sound/vibration levels as low as –60dB.

26. Who among the following gave the term 'Nano-technology' and when?

(a) Richard Feynman - 1959 (b) Norio Taniguchi - 1974
(c) Eric Drexler - 1986 (d) Sumiolijima - 1991

Uttarakhand P.C.S. (Pre) 2016

Ans. (b)

Norio Taniguchi was a professor at Tokyo University. He used the term 'Nanotechnology' for the first time in his research paper published in 1974.

27. With reference to carbon nanotubes, consider the following statements :

1. They can be used as carriers of drugs and antigens in the human body.
2. They can be made into artificial blood capillaries for an injured part of human body.
3. They can be used in biochemical sensors.
4. Carbon nanotubes are biodegradable.

Which of the statements given above are correct?

(a) 1 and 2 only (b) 2, 3 and 4 only
(c) 1, 3 and 4 only (d) 1, 2, 3 and 4

I.A.S. (Pre) 2020

Ans. (c)

Carbon nanotubes (CNTs) are allotropes of carbon made of graphite and constructed in cylindrical tubes with nanometre in diameter. CNTs have been successfully applied in pharmacy and medicine due to their high surface area that is capable of absorbing or conjugating with a wide variety of therapeutic and diagnostic agents (drugs, genes, vaccines, antibodies, biosensors, etc.). They have been first proven to be an excellent vehicle for drug delivery directly into cells without metabolism by the body. Hence, statement 1 is correct. NASA has successfully demonstrated biochemical sensors using carbon nanotube arrays. As per a September 2019 report of 'The Hindu', the Delhi - based researchers have fabricated a highly sensitive carbon nanotube-based sensor capable of detecting multidrug-resistant myeloid leukaemia cells. Thus, statement 3 is correct.
Statement 4 is also correct as multiple types of microbes and enzymes have the ability to biodegrade carbon nanotubes, graphene and their derivatives and in the future, more species with this ability will be found.
As per a research paper published in Science Daily in 2015, non-functionalised carbon nanotubes both soluble and surface-bound are not blood-compatible as scientists found that carbon nanotubes stimulate blood platelet activation, subsequently leading to serious and devastating blood clotting. Scientists are researching on carbon nanotubes and 3D bioprinting to create artificial blood vessels, but it is yet to be fully realized. So, statement 2 is incorrect in the present context.

28. **With reference to the use of nanotechnology in health sector, which of the following statements is/are correct?**

1. **Targeted drug delivery is made possible by nanotechnology.**
2. **Nanotechnology can largely contribute to gene therapy.**

Select the correct answer using the code given below.

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2015

Ans. (c)

Nanotechnology is a rapidly expanding area of research with huge potential in many sectors ranging from healthcare to construction and electronics. In medicine, it promises to revolutionize drug delivery, gene therapy, diagnostics and many areas of research, development and clinical application. Discovery of nanomedicine has given rise to nanoparticles through which better target specific drug and gene delivery is possible. Nanotechnology enables us to deliver the drug in the form of dendrimers, liposomes, nanoshells, emulsions, nanotubes, quantum dots etc. for the manipulation of various diseases and their metabolic pathway. It is of great importance in treatment and diagnosis of cancer. Hence, both given statements are correct.

29. **India's first nano-cellulose plant developed by ICAR-CIRCOT, Mumbai is related to :**

(a) Cotton (b) Jute
(c) Bamboo (d) Rice

Rajasthan P.C.S. (Pre) 2024

Ans. (a)

India's first nano-cellulose plant developed by ICAR-CIRCOT, Mumbai is related to cotton. This is first plant of its kind in India and unique in the world that can use cotton linters and cotton wastes as the raw material for preparation of nano-cellulose. The objective of nano-cellulose plant is to demonstrate ICAR-CIRCOT's technology to various stakeholders, product development, technology incubation and licensing.

30. **The most important property of nanomaterials is –**

(a) Force (b) Friction
(c) Pressure (d) Temperature

R.A.S./R.T.S.(Pre) 2013

Ans. (b)

Friction is the most important property of nanomaterials.

31. **If in a material one dimension is reduced to the Nano range while the other two dimensions remain large, the structure so obtained is known as :**

(a) Quantum step (b) Quantum well
(c) Quantum dot (d) Quantum wire

Rajasthan P.C.S. (Pre) 2023

Ans. (b)

If one dimension is reduced to the nano range while the other dimensions remain large, them we obtain a structure known as quantum well. If two dimensions are so reduced and one remains large, the resulting structure is referred to as a quantum wire. The extreme case of this process of size reduction in which all three dimensions reach the low nanometer range is called a quantum dot.

32. **A quantum dot is :**

(a) Electron microscopy image of nanostructures smaller than 1 nanometre
(b) Nanoscale analog of radio antennas
(c) A fictional nanorobot
(d) A semiconductor nanostructure

R.A.S./ R.T.S. (Pre) 2021

Ans. (d)

Quantum dots (QDs) are semiconductor nanostructures which exhibit size and composition-dependent optical and electronic (optoelectronic) properties. QDs are ultrasmall, typically falling in the size range between 1.5 and 10.0 nm. Recently, QD nanotechnology has successfully entered numerous electronic and biomedical industries. QDs have been demonstrated successfully due to their unique properties including superior photostability, size-dependent optical properties, high extinction coefficient and brightness, and large Stokes shift.

33. **Arrange the following products/examples of nano-technology in ascending order of the four generations of nanotechnology [I $\rightarrow$ IV] and select the correct answer using the codes given below :**

A. Aerosol
B. 3D networking
C. Molecular manufacturing
D. Targeted drugs

Code :

(a) D, A, B, C (b) D, A, C, B
(c) A, B, C, D (d) A, D, B, C

R.A.S./R.T.S. (Pre) 2016

Ans. (d)

The ascending order of the products/examples of four generations of Nanotechnology –

(I)	Products of first generation	–	Aerosols, Colloids, Polymers, Ceramics
(II)	Products of second generation	–	Targeted drugs, 3-D transistors, Amplifiers
(III)	Products of Third generation	–	Robotics, 3-D Networking
(IV)	Products of Fourth generation	–	Molecular manufacturing

From the above description it is clear that option (d) is the correct answer.

34. The 'Nano hummingbird' is –

(a) A new species of hitherto undiscovered hummingbird
(b) An extremely small electric car that can take 360^0 turn.
(c) A pocket sized unmanned spy plane developed in the U.S.A.
(d) A new variety of honey bee.

U.P.P.C.S. (Mains) 2010

Ans. (c)

The Nano Hummingbird or Nano Air Vehicle (NAV) is a tiny, remote controlled aircraft built to resemble and fly like a hummingbird, developed in the United States by AeroVironment Inc.

35. The first successful attempt to tap underground heat was made in :

(a) Boise, Idaho (USA) (b) Tel Aviv, Israel
(c) Tokyo, Japan (d) Canberra, Australia

M.P. P.C.S. (Pre) 2023

Ans. (a)

The first successful attempt (1890) to tap the underground heat was made in the city of Boise, Idaho (U.S.A.), where a hot water pipe network was built to give heat to the surrounding buildings. This plant is still working.

36. What is the location of the Enron Power Project?

(a) Kalol (b) Ahmednagar
(c) Virar (d) Dabhol

M.P.P.C.S. (Pre) 1996

Ans. (d)

The Dabhol Power Company (DPC) was formed to manage Dabhol Power Plant. It was built through the combined effort of Enron, GE and Bechtel. It is located at Dabhol Guhagar Taluka, Ratnagiri district (Maharashtra). At present, it is owned by Ratnagiri Gas and Power Pvt. Ltd., a subsidiary of NTPC Ltd.

37. Which of the following thermal power stations is situated at Chachai of Anuppur district?

(a) Satpura power station (b) Amarkantak power station
(c) Bansagar power station (d) None of the above

M.P. P.C.S. (Pre) 2023

Ans. (b)

Amarkantak thermal power station is situated at Chachai of Anuppur district, Madhya Pradesh. It is one of the coal-fired power station of M.P. Power Generation Company Limited (MPPGCL).

38. The theme of Indian Science Congress 2001 was :

(a) "Food nutrition and environmental security"
(b) "Arrest declining interest in pure sciences"
(c) "Make India energy self-sufficient"
(d) "Make India I.T. Superpower"

I.A.S. (Pre) 2001

Ans. (a)

The 88^{th} Indian Science Congress (ISC) conference was held between 3 to 7 January, 2001 in New Delhi. The Main theme of this conference was 'Food, Nutrition, and Environmental Security'. In January 2023, 108^{th} ISC was held at R.T.M. Nagpur University with the focal theme of 'Science and Technology for Sustainable Development with Women Empowerment'.

39. Which one of the following organizations won the CSIR Award for Science and Technology (S&T) Innovations for Rural Development, 2006 ?

(a) CLRI (b) IARI
(c) NDDB (d) NDRI

I.A.S. (Pre) 2007

Ans. (a)

On September 26, 2006 the then Prime Minister Dr. Manmohan Singh gave away the CSIR award for S&T Innovations for Rural Development to Central Leather Research Institute (CLRI), Chennai at Vigyan Bhawan, New Delhi.

40. Who among the following was awarded Padma Vibhushan - 2023 in the field of Science and Engineering?

(a) Shri S.M. Krishna
(b) Shri Dilip Mahalanabis

(c) Shri Srinivasa Varadhan
(d) Shri Narinder Singh Kapany

M.P. P.C.S. (Pre) 2023

Ans. (c)

Padma Vibushan-2023 in the field of Science and Engineering was awarded to Shri S.R. Srinivasa Varadhan of U.S.A. He is an Indian-American mathematician. He is known for his fundamental contributions to probability theory and in particular for creating a unified theory of large deviations. In the year 2007, he became the first Asian to win the Abel Prize.

41. Which one of the following organizations is not related to science and technology?

(a) DST (b) CSIR
(c) ICSSR (d) DAE

U.P. Lower Sub. (Pre) 2008

Ans. (c)

DST – Department of Science and Technology, Government of India
CSIR – Council of Scientific and Industrial Research
ICSSR – Indian Council of Social Science Research
DAE – Department of Atomic Energy, Government of India
Thus, ICSSR is not related to Science and Technology.

42. Which of the following statements is/ are correct regarding National Innovation Foundation–India (NIF) ?

1. NIF is an autonomous body of the Department of Science and Technology under the Central Government.

2. NIF is an initiative to strengthen the highly advanced scientific research in India's premier scientific institutions in collaboration with highly advanced foreign scientific institutions.

Select the correct answer using the code given below :

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2015

Ans. (a)

National Innovation Foundation-India (NIF) is an autonomous body of the Department of Science and Technology, Government of India. It started functioning in March, 2000 as India's national initiative to strengthen the grassroots technological innovations and outstanding traditional knowledge. Its mission is to help India become a creative and knowledge-based society by expanding policy and institutional space for grassroots technological innovators. Hence, only statement 1 is correct.

43. Match list-I with List- II and select the correct answer using the code given below the lists :

	List- I	List - II
A.	Chevron	1. Wind energy
B.	AT&T	2. Oil
C.	AMD	3. Telephone, Internet
D.	Enercon GmbH	4. Micro-processor

Code :

	A	B	C	D
(a)	2	1	4	3
(b)	4	3	2	1
(c)	2	3	4	1
(d)	4	1	2	3

I.A.S. (Pre) 2007

Ans. (c)

Chevron Corporation is an American multinational energy corporation. It is headquartered in San Roman, California and active in more than 180 countries. Chevron is engaged in every aspect of the oil, natural gas, and geothermal energy industries. AT&T is an American multinational telecommunications corporation headquartered at Whitacre Tower in Dallas, Texas. AMD (Advanced Micro Device) Inc. is an American multinational fabless semiconductor company based in Santa Clara, California, United States. AMD designs, develops and sells computer processors and related technologies for business and consumer markets. ENERCON GmbH is a German multinational wind turbine manufacturer company. It was established in 1984 by Aloys Wobben.

44. 'Dakshin Gangotri' is located in –

(a) Uttarakhand (b) Arctic
(c) Himalaya (d) Antarctica

48th to 52nd B.P.S.C. (Pre) 2005

Ans. (d)

Dakshin Gangotri, the first permanent research station of India was established in Antarctica. It was built up in 1984 and buried in 1990. MAITRI, India's second permanent research base in Antarctica was built and finished in 1989. BHARATI, India's third and newest permanent research base which is situated on a rocky promontory fringing the Prydz Bay between Storns and Broknes Peninsula in the Larsemann Hill area. It was commissioned on 18 March, 2012.

45. First Indian Station set up for Antarctic research is called :

(a) Dakshin Gangotri (b) Dakshini Yamunotri
(c) Antarctica (d) Godavari

U.P.P.C.S. (Mains) 2006

Ans. (a)

See the explanation of above question.

46. **The recent research station set up in the Antarctica is called :**
(a) Bharti (b) Dakshini Gangotri
(c) Maitri (d) None of the above

U.P.P.C.S. (Mains) 2009

Ans. (a)

See the explanation of above question.

47. **The name of the new research station being set up in Antarctica by India is :**
(a) Dakshin Gangotri (b) Bharti
(c) Dakshinayan (d) Maitri

U.P.P.S.C. (GIC) 2010

Ans. (b)

See the explanation of above question.

48. **The term 'IndARC', sometimes seen in the news, is the name of :**
(a) An indigenously developed radar system inducted into Indian Defence.
(b) India's satellite to provide services to the countries of Indian Ocean Rim.
(c) A scientific establishment set up by India in Antarctic region.
(d) India's underwater observatory to scientifically study the Arctic region.

I.A.S. (Pre) 2015

Ans. (d)

India's first multi-sensor underwater moored observatory 'IndARC' was successfully deployed in Kongsfjorden of the Arctic roughly halfway between the North Pole and Norway on July 23, 2014.

49. **When was the undersea cable-based Chennai-Andaman and Nicobar (CANI) Project inaugurated and dedicated to the nation by Prime Minister Shri Narendra Modi ?**
(a) August 15, 2022 (b) August 10, 2019
(c) January 26, 2023 (d) August 10, 2020

70th B.P.S.C. (Pre) 2024

Ans. (d)

Prime Minister Shri Narendra Modi on 10 August, 2020 launched and dedicated to the nation, the Chennai-Andaman & Nicobar Islands (CANI) submarine Optical Fibre Cable (OFC) connecting Andaman & Nicobar Islands to the mainland (Chennai) through video conferencing. The foundation stone for this project was laid by the PM on 30 December, 2018 at Port Blair. Earlier, A&N Islands were connected to the mainland through Satellite link with limited bandwidth capacity with high latency. Hence, the Government of India decided to install Submarine Optical link from the Mainland (Chennai) to A&N Islands to provide high speed internet connectivity.

50. **What is the name of the deep-sea submersible that imploded during an underwater voyage to the Titanic wreckage?**
(a) Alvin (b) Falcon
(c) Trident (d) Titan

69th B.P.S.C. (Pre) 2023

Ans. (d)

On 18 June 2023, Titan, a deep-sea submersible operated by the American tourism and expeditions company OceanGate, imploded during an expedition to view the wreck of the Titanic in the North Atlantic Ocean off the coast of New Foundland, Canada.

51. **Which one of the following statements best describes the 'Polar Code'?**
(a) It is the international code of safety for ships operating in polar waters.
(b) It is the agreement of the countries around the North Pole regarding the demarcation of their territories in the polar region.
(c) It is a set of norms to be followed by the countries whose scientists undertake research studies in the North Pole and South Pole.
(d) It is a trade and security agreement of the member countries of the Arctic Council.

I.A.S. (Pre) 2022

Ans. (a)

International Maritime Organization (IMO)'s International Code for Ships Operating in Polar Waters (Polar Code) is mandatory under both the International Convention for the Safety of Life at Sea (SOLAS) and the International Convention for the Prevention of Pollution from Ships (MARPOL). The Polar Code covers the full range of design, construction, equipment, operational, training, search and rescue and environmental protection matters relevant to ships operating in the inhospitable waters surrounding the two poles. The Polar Code entered into force on 1 January, 2017.

52. **At which place in Uttarakhand did the Indian Meteorological Department install a Doppler Weather Radar?**
(a) Dehradun (b) Chakrata
(c) Mukteshwar (d) Chamoli

Uttarakhand P.C.S. (Pre) 2021

Ans. (c)

Uttarakhand's first Doppler Weather Radar (DWR), a device which can read the intensity of precipitation in the air to predict severe weather conditions like storms and heavy rainfall, was inaugurated in Mukteshwar, Nainital district, in January, 2021. The radar has been installed under the Integrated Himalayan Meteorological Project (IHMP) of the Ministry of Earth Sciences.

53. **Which of the following is not correctly matched –**

(a) Isobar – Pressure Air
(b) Isoheight – Height
(c) Isohaline – Snowfall
(d) Isobath – Depth

U.P. U.D.A./L.D.A. (Pre) 2001

Ans. (c)

Isohaline is a line drawn on a map or chart to indicate connecting points of equal salinity in the ocean. Isobar is a line connecting points of equal atmospheric pressure. Isoheight is a line of constant height above a certain reference point, while Isobath line connects points of equal underwater depth.

54. **Which one of the following is correctly matched?**

(a) Ecocline – Biodiversity
(b) Pycnocline – Sea water density
(c) Thermocline – Air temperature
(d) Halocline – Sea water temperature

U.P.P.C.S. (Pre) 2024

Ans. (b)

Ecocline : It is a zone of gradual but continuous change from one ecosystem to another when there is no sharp boundary between the two in terms of species composition.
Thermocline : It is a layer in the ocean where there's a rapid change in temperature with depth. It separates the warmer, less dense surface water from the colder, denser deep water.
Halocline : It is a layer in the ocean where salinity changes rapidly with depth. It's often found below the well-mixed, uniformly saline surface water layer.
Pycnocline : It is a layer in the ocean where density changes rapidly with depth. Density is primarily determined by temperature and salinity, so the pycnocline often coincides with the thermocline and halocline.
Hence, pair of option (b) is correctly matched.

55. **Which one of the following shows density gradient in the body of water?**

(a) Ecocline (b) Halocline
(c) Pycnocline (d) Thermocline

U.P. P.C.S. (Mains) 2016

Ans. (c)

Pycnocline is the cline or layer which shows density gradient in the body of water. Halocline shows salinity gradient within a body of water whereas the Thermocline is a transition layer between deep and surface water (or mixed layer).

56. **"I am a citizen of milky way."**
The above statement is attributed by –

(a) Archana Sharma (b) Kalpana Chawala
(c) Satish Dhawan (d) Vikram Sarabhai

U.P.P.S.C. (GIC) 2010

Ans. (b)

'Kalpana Chawala' was an astronaut and space shuttle mission specialist of STS-107 (Columbia), who was killed when the craft disintegrated after re-entry into earth's atmosphere. "I am the citizen of milky way", this statement has been credited to Kalpana Chawala.

57. **The first heavy water plant was established in –**

(a) Bangalore (b) Bhopal
(c) Nangal (d) Hyderabad

U.P.P.S.C. (GIC) 2010

Ans. (c)

The Department of Atomic Energy (DAE) of Indian government commissioned the first heavy water plant at Nangal, Punjab in the premises of National Fertilisers Limited in 1962.

58. **Which one of the following is paramagnetic in nature?**

(a) Iron (b) Hydrogen
(c) Oxygen (d) Nitrogen

I.A.S. (Pre) 1997

Ans. (c)

Paramagnetism is a form of magnetism whereby certain materials are attracted by an externally applied magnetic field and form internal, induced magnetic fields in the direction of the applied magnetic field. Oxygen, Platinum, Sodium, Aluminium, Manganese, Potassium and Chromium are paramagnetic in nature.

59. **Which one of the following metal is not attracted by a magnet :**

(a) Iron (b) Nickel
(c) Cobalt (d) Aluminium

U.P.P.C.S. (Pre) 2002

Ans. (d)

Iron, Nickel, and Cobalt are ferromagnetic elements and quickly attracted towards the magnet. Aluminium is a paramagnetic element as its outermost electron is unpaired, so the induced magnetic field in Aluminium is low.

60. **Which of the following is non-electromagnetic element –**

(a) Nickel (b) Cobalt
(c) Chromium (d) Copper

U.P.P.C.S. (Pre) 1990

Ans. (d)

According to the options given in the question only Copper is a non-electromagnetic element.

61. **The magnetic needle points to –**

(a) East (b) West
(c) North (d) Sky

47th B.P.S.C. (Pre) 2005

Ans (c)

The magnetic needle points to north. All magnets have two poles, north pole and south pole. The north pole of one magnet is attracted to the south pole of the another magnet. The earth is a magnet that can interact with other magnets in this way. So the north end of a compass magnet is drawn to align with the earth's magnetic field. Because the earth's magnetic north pole attracts the 'north' ends of other magnets, it is technically the 'south pole' of our planet's magnetic field.

62. **With which of the following is the tape of the tape recorder coated?**

(a) Copper sulphate (b) Ferromagnetic powder
(c) Zinc oxide (d) Mercury

U.P.P.C.S. (Pre) 1998

Ans. (b)

The tape of a tape recorder consists of a plastic backing coated with a thin layer of tiny particles of ferromagnetic powder. The ferromagnetic powder is made up of small particles of iron. At the time of recording, they acquire magnetism through the particle.

63. **The World's most accurate clock that loses just a second every 300 million years uses –**

(a) Quartz atoms (b) Silicon atoms
(c) Strontium atoms (d) Zinc Atoms

U.P.P.C.S. (Mains) 2008

Ans. (c)

The world's most accurate atomic clock that loses just a second every 300 million years uses Strontium atoms. This clock is based on neutral atoms and has been demonstrated by physicists at JILA (Joint Institute of Laboratory of Astrophysics) a Joint Physics Institute of the National Institute of Standards and Technology (NIST) and the University of Colorado of USA. But recently the JILA has also developed a new atomic clock that will not lose or gain a second in 15 billion years. The scientists claim that it is the most accurate clock ever built.

64. **In automatic wrist watches energy is provided by –**

(a) Manual binding (b) Battery
(c) Liquid crystal (d) Different movements of our hand

U.P.P.C.S. (Pre) 2007

Ans. (d)

The automatic watches do not require batteries for power to run. They obtained their energy from the movement of the wearer's hand. If these watches will not be worn for 2 to 3 days by a person, they will stop.

65. **In an electronic watch, the component corresponding to pendulum of a pendulum clock is a :**

(a) Transistor (b) Crystal oscillator
(c) Diode (d) Balance wheel

I.A.S. (Pre) 1994

Ans. (b)

In an electronic watch 'Crystal Oscillator' is used as its time-keeping element, while the pendulum clock uses a pendulum that is a swinging weight as its timekeeping element. A crystal oscillator uses the inverse piezoelectric effect to convert vibrations into stable oscillations. It works by applying an alternating voltage to a crystal, causing it to vibrate at its natural frequency.

66. **The working of the quartz crystal in the watch is based on the :**

(a) Photoelectric effect (b) Johnson effect
(c) Piezoelectric effect (d) Edison effect

I.A.S. (Pre) 1993

Ans. (c)

The working of the quartz crystal in watch is based on piezoelectric effect. The piezoelectric effect is the ability of certain materials to generate an electric charge in response to applied mechanical stress.

67. **Full form of MOEMS is :**

(a) Micro-Optic Electronic Media Source
(b) Micro-Opto-Electro-Mechanical-Systems
(c) Mega Operations Electronic Media Software
(d) Micro-Optic-Electro-Mechanical Source

R.A.S./ R.T.S. (Pre) 2021

Ans. (b)

Full form of MOEMS is 'Micro-Opto-Electro-Mechanical-Systems'. MOEMS, also known as optical MEMs, are integrations of mechanical, optical, and electrical systems that involves sensing or manipulating optical signals at a very small size. MOEMS includes a wide variety of devices (e.g. optical switch, microbolometers, tunable VCSEL etc.), which are usually fabricated using micro-optics and standard micromatching technologies using materials like silicon, silicon dioxide and gallium arsenide.

68. **What does 'PUMA' stands in context of Robotics?**

(a) Programmable Used Machine to Assemble
(b) Programmed Utility Machine for Assembly

(c) Programmable Universal Machine for Assembly
(d) Programmed Utility Machine to Assemble

M.P. P.C.S. (Pre) 2020

Ans. (c)

Programmable Universal Machine for Assembly (PUMA) or Programmable Universal Manipulation Arm (PUMA) is an industrial robot arm developed by Victor Scheinman at the pioneering robot company Unimation. Initially developed for General Motors, the PUMA was based on earlier designs Scheinman invented while at Stanford University, USA.

69. Consider the following activities:

1. Identification of narcotics on passengers at airports or in aircraft

2. Monitoring of precipitation

3. Tracking the migration of animals

In how many of the above activities can the radars be used?

(a) Only one (b) Only two
(c) All three (d) None

I.A.S. (Pre) 2024

Ans. (b)

Radar, or radio detection and ranging, uses radio waves to detect and locate objects by transmitting pulses and analyzing the reflected signals. Radars today are used to detect and track aircraft, spacecraft, and ships at sea as well as insects and birds in the atmosphere; measure the speed of automobiles; map the surface of the Earth from space; and measure properties of the atmosphere and oceans.

Weather radar, also called weather surveillance radar (WSR) and Doppler weather radar, is a type of radar used to locate precipitation, calculate its motion, and estimate its type (rain, snow, hail etc.). Hence, point 2 is correct.

Radar can be used to track the migration of animals such as birds, bats, and insects. The NEXRAD Doppler system is one type of radar that can determine the speed and direction of migrating animals, whether they are moving towards or away from the radar. Hence, point 3 is correct.

Radar is not used for identification of narcotics on passengers at airports or in aircraft. Trace detectors for narcotics are handheld electronic devices that use spectroscopic methods of detection to identify a range of illegal drugs. Hence, point 1 is incorrect.

Thus, option (b) is the correct answer.

70. Consider the following activities :

1. Spraying pesticides on a crop field

2. Inspecting the craters of active volcanoes

3. Collecting breath samples from spouting whales for DNA analysis

At the present level of technology, which of the above activities can be successfully carried out by using drones?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2020

Ans. (d)

At the present level of technology, drones can be used for spraying pesticides on a crop field, for inspecting the craters of active volcanoes and also for collecting breath samples from spouting whales for DNA analysis. As per an article of December, 2019 farmers of Khammam district (Telangana) are hiring drones to spray pesticides in their fields. Drones can capture images of erupting volcano craters that would be impossible to get otherwise. Researchers are already using drones at volcanoes in Indonesia and Guatemala, etc. In 2017, for the first time, scientists successfully used drone to gather samples of the fluids exhaled by spouting humpback whales for DNA analysis and tracking their health.

71. When and where was the Central Electrochemical Research Institute established with the efforts of Alagappa Chettiar, Dr. Shanti Swaroop Bhatnagar and Pt. Jawahar Lal Nehru?

(a) Lucknow, 1951
(b) Karaikudi, 1953
(c) Chennai, 1948
(d) Shivganga, 1953

R.A.S./ R.T.S. (Pre) 2021

Ans. (b)

Central Electrochemical Research Institute (CECRI), founded in 1948, has its roots in the patriotic fervor of Dr. R.M. Alagappa Chettiar, Pandit Jawaharlal Nehru and Dr. Shanthi Swarup Bhatnagar. On January 14, 1953 CECRI became a physical reality when Dr. S. Radhakrishnan dedicated CECRI in Karaikudi (Tamil Nadu), the twelfth national laboratory under the CSIR, to the nation. CECRI represents the largest research establishment for electrochemistry in South Asia, Headquartered at Karaikudi, CECRI has extension centers in Chennai, Mandapam and Tuticorin.

72. When and with whose efforts was Indian Institute of Science established in Bangalore?

(a) 1917, Prafulla Chandra Ray
(b) 1930, J.C. Bose
(c) 1909, Jamshed ji Tata
(d) 1911, Meghnad Saha

R.A.S./ R.T.S. (Pre) 2021

Ans. (c)

The Indian Institute of Science (IISc) is a public, deemed, research university for higher education and research in science, engineering, design, and management. It is located in Bengaluru, in the Indian state of Karnataka. The institute was established in 1909 with active support from Jamshedji Tata.

73. Which of these companies is established by Government of Rajasthan with the aim to regulate developing needs of Information Technology skills in 21st century?

(a) Rajasthan Knowledge Company Limited
(b) Rajasthan Technology Private Limited
(c) Rajasthan Knowledge Corporation Limited
(d) Rajasthan Technology Corporation Limited

Rajasthan P.C.S. (Pre) 2023

Ans. (c)

The RKCL – Rajasthan Knowledge Corporation Limited is a Public Limited Company established by the Government of Rajasthan. The main objective of company is to develop a new educational framework which can plan, implement, supervise and regulate the developing needs for IT skills in the 21st century in the State of Rajasthan.

74. The National Centre for Earth Science Studies (NCESS) is situated in which State?

(a) Andhra Pradesh (b) Kerala
(c) Tamil Nadu (d) Karnataka

M.P. P.C.S. (Pre) 2025

Ans. (b)

The National Centre for Earth Science Studies (NCESS) is an autonomous research centre to promote scientific and technological research and development studies in the earth sciences. NCESS pursues problems related to land, sea and atmosphere. It was instituted by the government of Kerala as CESS in 1978, at Thiruvananthapuram, Kerala. CESS was the earliest institute in the country to embrace the concept of Earth System Science (ESS). At present, it is an autonomous institute under the Earth System Science Organization (ESSO) of Ministry of Earth Sciences, Government of India.

75. Which one of the following is not suitably matched ?

(a) National institute of Oceanography – Goa
(b) Indian National Centre for Ocean Information Services – Hyderabad
(c) National Institute of Ocean Technology – Chennai
(d) Antarctic Study Centre – Bangalore

U.P.P.C.S. (Mains) 2004

Ans. (d)

The National Centre for Antarctic and Ocean Research (NCAOR) is an Indian Research and Development Institution, situated at Vasco de Gama, Goa not in Bengaluru. Rest are correctly matched.

76. Which of the following is not correctly matched?

(a) Aryabhatta Research Institute of Observational Sciences – Nainital
(b) College of Defence Management – Secunderabad
(c) Central Institute of Indian Languages – Mumbai
(d) Indian Institute of Science – Bengaluru

M.P. P.C.S. (Pre) 2023

Ans. (c)

The Central Institute of Indian Languages is an Indian research and teaching institute based in Mysuru, not in Mumbai. Other given pairs are correctly matched.

77. Match List-I with List-II and select the correct answer using the codes given below in the lists.

List- I (Institute)
A. Central Institute of Higher Tibetan Studies
B. Indira Gandhi Institute of Development Research
C. National Institute of Mental Health and Neuro Sciences
D. Central Institute of English and Foreign Languages

List- II (Location)
1. Hyderabad **2. Mumbai**
3. Bangalore **4. Dharamshala**
5. Varanasi

Code :

	A	B	C	D
(a)	5	3	4	1
(b)	5	2	3	1
(c)	3	2	4	5
(d)	4	5	1	2

I.A.S. (Pre) 2000

Ans. (b)

Central Institute of Higher Tibetan Studies – Varanasi
Indira Gandhi Institute of Development Research – Mumbai
National Institute of Mental Health and Neuro Sciences – Bengaluru
Central Institute of English and Foreign Languages (Present name : The English and Foreign Languages University) – Hyderabad

78. 'Indian Institute of Naturopathy and Yogic Science' is located at :

(a) Pune (b) Lucknow

(c) Hyderabad (d) Bangalore

U.P.P.C.S. (Pre) 2002

Ans. (d)

Indian Institute of Naturopathy and Yogic Science is located at Bangaluru (Karnataka).

79. Match List- I with List- II and select the correct answer using the code given below :

List-I	List-II
A. Hi-Tec City	**1. Lucknow**
B. Science City	**2. Thumba**
C. Rocket Launching Centre	**3. Calcutta**
D. Central Drug Research Institute	**4. Hyderabad**

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	4	3	2	1
(c)	2	3	4	1
(d)	1	4	2	3

U.P.P.C.S. (Pre) 1999

Ans. (b)

The correctly matched lists are as follows :

Central Drug Research Institute	– Lucknow
Rocket launching Centre	– Thumba (Thiruvananthapuram)
Science City	– Calcutta
Hi-Tec city	– Hyderabad

80. A carbon microphone is best used in –

(a) Dynamo (b) Telephone
(c) Transformer (d) None of these

U.P.P.C.S. (Mains) 2009

Ans. (b)

A microphone is acoustic to electric transducer or sensor that converts sound into an electrical signal. The carbon microphone was used in early telephone repeaters, making long distance phone calls possible in the era before vacuum tube amplifiers. Carbon microphones can be used as amplifiers.

81. The laws of planetary motion were enunciated by –

(a) Newton (b) Kepler
(c) Galileo (d) Copernicus

U.P.P.S.C. (GIC) 2010

Ans. (b)

The three laws of planetary motions were proposed by Johannes Kepler in early 1600s. Kepler made it clear that every planet revolves around the Sun in an elliptical orbit.

82. Soleckshaw is a –

(a) Computer Program (b) Moon Buggy
(c) Soft sole of a footwear (d) Solar rickshaw

U.P.P.C.S. (Mains) 2008

Ans. (d)

Soleckshaw is an eco-friendly tricycle. It is driven partly by pedal and partly by electric power, supplied by a battery that is charged from solar energy. It has been developed by a team of scientists at CSIR national laboratory.

83. Bibliometry is :

(a) Function of Library Network
(b) Information Management Service
(c) Information Management Tool
(d) Library Service

U.P.P.C.S. (Pre) 2019

Ans. (c)

Bibliometry is an Information Management Tool. Bibliometry is a quantitative statistical technique to measure levels of production and dissemination of knowledge, as well as a useful tool to track the development of an scientific area.

84. Cytotron is the device by which is produced :

(a) Electrical energy (b) Artificial climate
(c) Sound (d) Picture on the screen

U.P. P.C.S. (Mains) 2016

Ans. (b)

Cytotron is the device by which artificial climate is produced. Cytotron is also the trade name given to a device that uses rotational field quantum nuclear magnetic resonance (RFQMR) which has been developed by the Centre for Advanced Research and Development (CARD), a division of Scalene Cybernetics in Bengaluru. It is a new device used in regenerative and degenerative tissue engineering and repairing. The cytotron is now being used for treating diseases like osteoarthritis and cancer.

85. Where was the 1st Engineering College of Asia established ?

(a) Chennai (b) Bangalore
(c) Roorkee (d) None of the above

Uttarakhand P.C.S. (Mains) 2006

Ans. (c)

The Roorkee College was established in 1847 AD as the first engineering college in the British Empire. The College was established in former Uttar Pradesh and now Uttarakhand. The college was renamed as Thomson College of Civil Engineering in 1854. It was given the status of University by

Act No. IX of 1948 of the United Province (Uttar Pradesh) in recognition of its performance and its potential and keeping in view the needs of post-independent India.

86. When the metric system was introduced in India –

(a) 1-10-1958 (b) 2-10-1956

(c) 1-4-1957 (d) 1-1-1958

M.P.P.C.S. (Pre) 1999

Ans. (c)

The metric system (decimal system) in India was introduced on April 1, 1957. The coins minted between 1957 to 1964 called 'Naya Paisa' (New Paisa).

87. A tachyon stands for –

(a) A particle moving faster than the velocity of light

(b) A constituent of heavier atomic nuclei

(c) A particle moving greater than the velocity of sound in air

(d) A quantum of lattice vibration

Uttarakhand P.C.S. (Pre) 2007

Ans. (a)

Tachyon is a Greek word which means rapid. Tachyon is a hypothetical particle that travels faster than light.

88. Match List-I with List-II and select the correct answer by using the codes given below the lists.

List-I (Person)	List-II (Known As)
A. John C. Mather	**1. Co-founder of Microsoft**
B. Michael Griffin	**2. Space Walker**
C. Paul G. Allen	**3. Administrator of NASA**
D. Piers Sellers	**4. Nobel Prize Winner, 2006 in Physics**

Code :

	A	B	C	D
(a)	4	1	3	2
(b)	2	3	1	4
(c)	4	3	1	2
(d)	2	1	3	4

I.A.S. (Pre) 2007

Ans. (c)

John C. Mather is a senior astrophysicist in the Observational Cosmology Laboratory located at NASA, Greenbelt. He was awarded 2006 Nobel Prize for physics, jointly with George F. Smoot. Michael Griffin served as administrator of NASA from April 13, 2005 to January 20, 2009. Paul G. Allen was the co-founder of Microsoft. He associated with Bill Gates in 1975. Piers J. Sellers was selected as an astronaut candidate by NASA in April 1996. He had logged a total of 34 days, 23 hours, 03 minutes and 56 seconds in space including almost 4 EVA hours in six space walks.

89. Consider the followings about Raja Ramanna :

1. He had directed the team of scientists which carried out the test of the nuclear device.

2. He was awarded Padma Vibhushan in 1975.

3. He was made Union Minister of State for Defence in 1990.

4. He had written a book entitled 'The Structure of Music in Raga and Western System'.

Select the correct answer by using the code given below:

(a) 1, 2, 3 and 4 (b) 1 and 2

(c) 1, 2 and 3 (d) 4 only

R.A.S./R.T.S. (Pre) 2013

Ans. (a)

Raja Ramanna was a multifaceted personality. An eminent nuclear physicist, a highly accomplished technologist, an able administrator and a gifted musician. He had directed the team of scientists which carried out the India's first peaceful nuclear experiment in Rajasthan. He was awarded Padma Vibhusana Award in 1975. He also served as the Minister of State for Defence in the Union Cabinet (January to November 1990). The book 'The structure of music in Raga and Western System' (1993) was written by him.

90. Water Jet Technology finds application in –

(a) Irrigation (b) Drilling of mines

(c) Firefighting (d) Mob control

U.P.P.C.S. (Mains) 2004

U.P.P.C.S. (Pre) 1998

Ans. (b)

A Water Jet Technology is capable of cutting a wide variety of materials using a very high-pressure jet of water or a mixture of water and an abrasive substance. This technology is mostly used in the drilling of mines and in aeronautics.

91. The laser beam is used for :

(a) Treatment of cancer (b) Treatment of heart

(c) Treatment of eye (d) Treatment of kidney

U.P.P.C.S. (Pre) 2002

Ans. (*)

The laser beam is used for the treatment of many diseases. This technology offers surgeon the ability to work precisely. Laser therapy is used in many procedures such as to shrink or destroy tumors, polyps or precancerous growths, relieve symptoms of cancer, remove part of the prostate, removing of kidney stones, repair a detached retina, improve vision etc.

92. **Match List-I with List-II and select the correct answer from the code given below :**

	List-I		List-II
A.	**Dry Ice**	1.	**Treatment of Cancer**
B.	**Gene therapy**	2.	**Freezing living bodies to be revived later**
C.	**Cryonics**	3.	**Solid carbon dioxide**
D.	**Cobalt-60**	4.	**Treatment of blood diseases.**

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	4	3	2	1
(c)	3	4	2	1
(d)	2	1	4	3

U.P.P.C.S. (Pre) 1997
U.P. Lower Sub. (Pre) 2004

Ans. (c)

Dry Ice, sometimes referred to as 'Cardice', is basically the solid form of carbon dioxide. Cobalt-60 is used for treatment of cancer. Cryonics is the procedure of freezing bodies of animals and humans with the hope that healing and resuscitation may be possible in the future. Gene therapy is an experimental technique that uses genes to treat or prevent disease.

93. **Which one of the following is not correctly matched ?**

(a) Y2K – Computer
(b) Arthritis – Uric Acid
(c) Noise Pollution – Decibel
(d) Adobe – Hardware

Uttarakhand Lower Sub. (Pre) 2010

Ans. (d)

'Adobe System' is an American multinational computer software company mainly focused on the creation of multimedia and software products. Thus it is clear that option (d) is not correctly matched.

94. **Which of the following statements is true?**

(a) Johns Hopkins was the founder of Homeopathy.
(b) Isaac Newton had propounded the Theory of Evolution.
(c) Carbon monoxide creates more air pollution.
(d) Vasco-de-Gama discovered America.

Chhattisgarh P.C.S. (Pre) 2008

Ans. (c)

Johns Hopkins was born on May 19, 1795 in Maryland. He was the founder of 'John Hopkins University' and John Hopkins Hospital. Dr. Samuel Hahnemann was the actual founder of Homeopathy. Charles Darwin, who was born in Shrewsbury, Shrophire, England, on 12 February 1809, propounded the 'Theory of Evolution'. Carbon monoxide is not absorbed by the plants. Hence, it creates more air pollution. Vasco-de-Gama was a Portuguese explorer. He was the first European to reach India by sea, while America was discovered by Christopher Columbus.

95. **Otto Hahn discovered atom bomb by the principle of –**

(a) Uranium fission (b) Nuclear fission
(c) Alpha radiation (d) Gamma radiation

Uttarakhand P.C.S. (Pre) 2006

Ans. (b)

Otto Hahn discovered atom bomb by the principle of Nuclear Fission. In 1939, the German scientists Otto Hahn and F. Strassmann determined that if the slow moving neutrons are bombarded on Uranium-235, then the nucleus of 235 breaks into two equal parts and releases lots of energy. This process is called nuclear fission.

96. **The principle of atom bomb is based on –**

(a) Nuclear fission (b) Nuclear fusion
(c) Nuclear spallation (d) None of these

U.P. U.D.A./L.D.A. (Pre) 2007

Ans. (a)

See the explanation of above question.

97. **What was the fissionable material used in the bombs dropped at Nagasaki (Japan) in the year 1945 ?**

(a) Sodium (b) Potassium
(c) Plutonium (d) Uranium

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

'Fat Man' was the code name for the type of bomb which was dropped on the Japanese city of Nagasaki on 9 August, 1945 by the United States of America. Plutonium was used as fissionable material in this bomb. On the other hand 'Little Boy' which was dropped on the Japanese city of Hiroshima on 6 August, 1945 used Uranium as fissionable material.

98. **Which country has test-launched first 3D television broadcast?**

(a) U.K. (b) China
(c) America (d) South Africa

M.P.P.C.S. (Pre) 2013

Ans. (c)

On April 29, 1953, Los Angeles television station KECA broadcasted an episode of the sci-fi program Space Patrol in 3D. The event was a technology demonstration for the 31st Annual National Television and Radio Broadcasters Convention, which was held in Los Angeles that week. Viewers were required to wear special 3D glasses created by Polaroid; otherwise, the show appeared as an indistinct blur. It was the first 3D television broadcast in the world.

99. Consider the following statements :

1. The most important technological application of liquid crystals is in digital display.

2. Modem is a device that is connected to a computer and to a phone line.

3. The National Institute of Oceanography is located at Coimbatore.

4. Virginis-70 is a system for recording video programmes.

Of these statements :

(a) Only 1 and 2 are correct

(b) Only 2 and 3 are correct

(c) Only 2, 3 and 4 are correct

(d) Only 3 and 4 are correct

U.P.P.C.S. (Mains) 2009

Ans. (a)

A Liquid Crystal Display (LCD) is a flat panel display which is used to display the text, images, video etc. by the electronic method. A modem modulates outgoing digital device to analog signals for a conventional copper twisted pair telephone line and demodulates the incoming analog signal and converts it to a digital signal for the digital device. It is a device that is connected to a computer and to a phone line. The headquarter of National Institute of Oceanography is located in Goa while Regional centres are in Kochi, Mumbai and Visakhapatnam. Virginis-70 is a yellow dwarf star approximately 59 light-years away in the constellation Virgo.

100. Liquid crystals are used in :

(a) Wrist Watches

(b) Display Devices

(c) Pocket Calculators

(d) All of the above

U.P.P.C.S. (Pre) 1996

Ans. (d)

Liquid crystals are used in wrist watches, display devices, pocket calculators and in many portable computers.

101. Organic Light Emitting Diodes (OLEDs) are used to create digital display in many devices. What are the advantages of OLED displays over Liquid Crystal displays?

1. OLED displays can be fabricated on flexible plastic substrates.

2. Roll-up displays embedded in clothing can be made using OLEDs.

3. Transparent displays are possible using OLEDs.

Select the correct answer using the code given below:

(a) 1 and 3 only

(b) 2 only

(c) 1, 2 and 3

(d) None of the above statements is correct

I.A.S. (Pre) 2017

Ans. (c)

An organic light-emitting diode (OLED) is a light-emitting diode (LED) in which the emissive electroluminescent layer is a film of organic compound that emits light in response to an electric current. OLED displays can be fabricated on flexible plastic substrates. Thus statement 1 is correct. Roll-up displays embedded in clothing can be made using OLEDs. Transparent displays are also possible using OLEDs. Hence, statement 2 and 3 are also correct.

102. With a 16 : 9 picture aspect ratio, display resolution 1080p means :

(a) 1080 × 1080 pixels

(b) 1920 × 1080 pixels

(c) 720 × 1080 pixels

(d) 3840 × 1080 pixels

R.A.S./ R.T.S. (Pre) 2021

Ans. (b)

With a 16 : 9 picture aspect ratio, display resolution 1080p means 1920 × 1080 pixels. It is usually known as FHD or 'Full HD' resolution. 1080p is a set of HDTV high-definition video modes characterized by 1920 pixels displayed across the screen horizontally and 1080 pixels down the screen vertically.

103. Which one of the following devices is used to cool the engine of the vehicles?

(a) Polygraph

(b) Turbine

(c) Radiator

(d) Quadrant

U.P.P.C.S. (Mains) 2014

Ans. (c)

Radiators are heat exchangers used for cooling internal combustion engines mainly in automobiles but also in piston-engine aircraft, railway locomotives, motorcycles, stationary generating plant or any similar use of such an engine.

104. Mobiles and Automobiles have brought about a revolution in the social life of Indians, especially in the rural, in terms of ?

1. Mobility of the people

2. Connectivity of the people

3. Sensitivity of the people

Select the correct answer from the code given below :

(a) 1 and 2 only

(b) 1 and 3 only

(c) 2 and 3 only

(d) 1, 2 and 3

U.P. U.D.A./L.D.A. (Spl.) (Pre) 2010

Ans. (a)

Mobiles and Automobiles have brought about a great revolution in the social life of Indians, where mobiles help to connect the people and automobile help in mobilizing of the people.

105. Name of first 3d printed humanoid robot in India is:

(a) Indro (b) Robcop

(c) Mitra (d) Manav

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (d)

'Manav' is a humanoid robot developed in the laboratory of A-SET Training and Research Institutes by Diwakar Vaish (Head of Robotics and Research, A-SET Training and Research Institutes) in late December 2014. It debuted at the IIT-Bombay Techfest 2014-2015 in Mumbai. It's a 2-foot tall, 2-kilogram robot designed for research purposes. Manav has vision and sound processing capabilities and can perform tasks like walking, talking and even dancing. It is built with 21 sensors, two cameras in its eye sockets and two mikes on either side of its head.

106. What is the name of Tesla's humanoid robot launched in October, 2022?

(a) Sophia (b) Atlas

(c) Pepper (d) Optimus

69th B.P.S.C. (Pre) 2023

Ans. (d)

Tesla CEO Elon Musk revealed the prototype of its humanoid robot 'Optimus' in October, 2022. It was said that Optimus shares some Al software and sensors with its Tesla cars' Autopilot driver assistance features. Optimus has the ability to self-calibrate its arms and legs and can locate its limbs in space using vision and joint position encoders.

107. Which of the following is not an advantage of Robots?

(a) They can assist humans with disabilities.

(b) They can replace jobs.

(c) They can be used in dangerous environments.

(d) They do not get tired or require a break.

M.P. P.C.S. (Pre) 2022

Ans. (b)

One of the most significant disadvantages of robots, is that the use of robots can lead to job losses for human workers, particularly those in low-skill, repetitive jobs. Thus, this is not an advantage of robots. Other three given statements are the advantages of robots.

108. Which one of the following not-for-profit organization is the apex body for setting up standards in Robotics and Automation in India?

(a) All India Council for Robotics and Automation (AICRA)

(b) National Institute of Robotics and Artificial Intelligence (NIRA)

(c) India Science, Technology, Engineering and Maths (STEM) Mission

(d) Robotics Skill Centre of India (RSCI)

M.P. P.C.S. (Pre) 2025

Ans. (a)

All India Council for Robotics & Automation (AICRA), a not-for-profit organization, is the apex body for setting up standards in Robotics and Automation in India. Established in 2014 and ever since, AICRA's relentless pursuit has been to constantly support the Robotics & Automation industry, providing support systems to institutions such as quality assurance, information systems and train-the-trainer (TTT) academies either directly or through partnerships. AICRA is focused on building the architecture integral to the development of the automation sector through policy advocacy, and help in setting up the strategic direction for the sector to unleash its potential and dominate newer frontiers.